

EE 435

CHAPTER TWO:

Introduction to Data Communications:

2.0 Introduction

This chapter provides an overview of the telecommunication system's features. Whereas these features apply to every wireless transmission connection, we'll go into the specifics of short-range devices in greater depth.

Figure 2.1 is a simplified block design of a digital wireless connection. The connection transports data from one place, referred to as data sources, to some other site, termed reconstruction data.

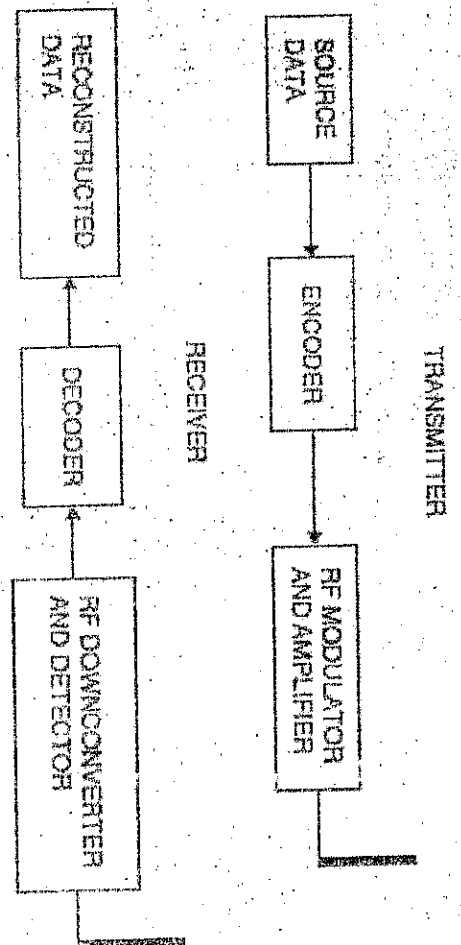


Figure 2.1 Diagram of radio transmission links

2.1 Data Communication Fundamentals

Data consists of information that is kept in digital form in Transmitting Data. The practice of exchanging electronic information among two or more sites is known as data communication. The term "information" refers to knowledge or intellect. The broadcast, receiving, and analysis of digital data may be summed up as data communications. The technology and communication have to be part of a communication network that involves the combination of hardware (physical equipment) and software for data transmission to take place (programs). A digital communications system's performance can be determined by four key features: transmission, precision, reliability, and jitter.

A digital communications system consists of five parts:

1. **Message:** The information (data) to also be delivered is represented by the message. Text, numbers, photos, music, and video are all common types of information.
2. **Sender:** The equipment that delivers the data packet is referred to as the sender. It might be a laptop, desktop, telephone wireless device, video camera, or another device.
3. **Receiver:** The equipment that receives information is referred to as the receiver. It might be a pc, desktop, phone handset, tv, or other devices.
4. **Transmission medium:** The tangible path that a message takes from the source node to the destination node is referred to as the transmission channel. Twisted-

pair cable, coaxial cable, fiber-optic cable, and microwaves are all examples of transmission mediums.

5. **Protocol:** A protocol is a collection of rules that control the transmission of data. It is a contract between both the communicating entities.

2.2 Data Communication Standards Organizations

Standards organizations are made up of organizations, governments, manufacturers, and users that are instrumental in creating, organizing, and keeping order. The goal is for all producers and consumers of data communication systems to adhere to these guidelines. The following are the key data communication standards for organizations:

2.2.1 International Standard Organization (ISO)

ISO is a global standardized organization that covers a wide variety of topics. It is mostly made up of representatives of various countries' standards committees from across the globe. It's also in charge of creating designs that deliver increased rates of system interoperability, quality enhancement, higher productivity, and cost reductions. The ISO is indeed in charge of approving and coordinating the activities of other standards bodies.

2.2.2 International Telecommunications Union (ITU-T)

The International Telecommunications Union (ITU-T) is among the four large components of the International Telecommunications Union, which is headquartered in Geneva, Switzerland. Three kinds of requirements have been created: The V series is designed for modem interface and data transmission over telephone lines, whereas the X series is designed for data transfer via open digital networks, email, and directory services. The I and Q series for Integrated Services Digital Network (ISDN) and its expansion Broadband ISDN. The ITU-T membership comprises government officials and delegates from numerous nations, and it is the current United Nations standardization agency.

2.2.3 Institute of Electrical and Electronics Engineers (IEEE)

IEEE is a global professional body comprised of electronics, computer, and communications engineers that were created in the U. S. Including over 1 million representatives, it is now the world's greatest professional body. It creates information and communication processing guidelines with the overarching purpose of developing the concept, inventiveness, and quality of the product in any electrical engineering field.

2.2.4 American National Standards Institute (ANSI)

The American National Standards Institute (ANSI) is the authorized standards organization of the United States or the United States' voting representation to the ISO. ANSI is a wholly autonomous, quasi-association formed of information processing hardware manufacturers and users. Members of ANSI come from professional organizations, industrial groups, government regulatory entities, and consumer products manufacturers.

2.2.5 Electronics Industry Association (EIA)

The EIA is a non-profit business group in the United States that develops and promotes engineering specifications. EIA operations comprise standard production, information

dissemination, and advocacy, and it is in charge of producing the RS (recommended standard) family of computer and communications specifications.

2.2.6 Telecommunications Industry Association (TIA)

The TIA is the industry's foremost trade organization for information and communications technology. It promotes company growth prospects via business expansion, commercial marketing, industry exhibitions, and the creation of standards. It comprises producers of information and communications technology devices and supports the confluence of emerging communications infrastructure.

2.2.7 Internet Architecture Board (IAB)

IAB, formerly known as the Internet Operations Board, is a body established by ARPA (Advanced Research Projects Agency) to review ARPANET operations with the goal of accelerating the creation of technology relevant to the American Army. The Internet Association Board (IAB) is the Internet Society's technical advisory board, and its tasks include:

- I. Supervises the World wide web architectural protocols and processes.
- II. Controls the methods being used to produce Internet Standards and functions as an appeals board for objections about incorrect standardization process implementation.
- III. In charge of managing the numerous Internet-assigned identities

2.2.8 Internet Engineering Task Force (IETF)

The Internet Engineering Task Force (IETF) is a huge multinational society of network engineers, administrators, suppliers, and academics interested in the growth of Internet infrastructure and its seamless functioning.

2.2.9 Internet Research Task Force (IRTF)

The IRTF encourages scientific studies for the emergence of the future Internet by forming concentrated, long-term, and focused research groups that concentrate on problems relating to HTTP protocol, programs, structure, and technologies.

2.3 Architecture of Layered Networks *HTTP, peer, transport protocols*

Typically, networks are arranged as multilayer networks or levels, each and everyone building on the one underneath it to decrease model complexity. The fundamental concept of hierarchical architecture is to break the design into little sections. Every layer augments the services supplied by the previous levels so that the top layer has a complete collection of functions to handle connections and operate applications. The advantages of layered models include simplicity and explicit interconnections, i.e., transparent architecture and comparison between elements of various suppliers. A fundamental idea is to establish layer autonomy by identifying services supplied from each layer to the next higher layer not specifying how the services will be implemented. This allows you to make changes to one layer without impacting the others. Services, protocols, and interfaces are the fundamental components of a layered model. A service is a collection of operations provided by one layer to the next (higher) layer. A *protocol* is a collection of rules used by a layer to communicate with a peer entity. These

guidelines apply to both the contents as well as the sequence in which the signals are used. Communication connections are defined between the layers. Information from one layer to the next is routed through these interfaces.

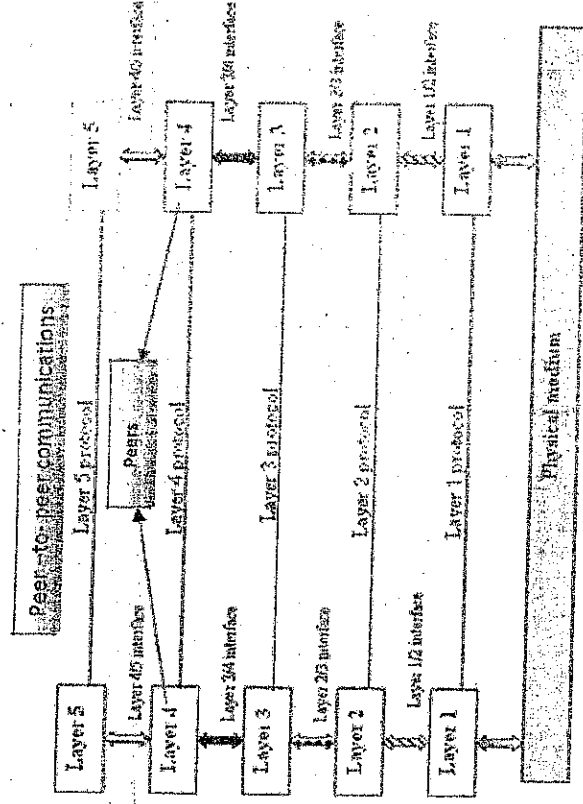


Figure 2.1 Peer-to-Peer Communication

Communication between two related levels in layered systems necessitates the use of a data packet known as a *Protocol Data Unit (PDU)*. A PDU could be used as a header at the start of a message or as a trailer at the conclusion of a message. The data in the source system goes downward via the layers and upwards at the destination node.

2.4 Open Systems Interconnection (OSI)

Around 1976, the International Standard Organization (ISO) formed a group to create communication and computer design, and the Network model is the product of that endeavor. The Open Systems Interconnection (OSI) conceptual framework for communications architecture was adopted as a global standard in 1985. The word "open" relates to the capability to link any two different systems that follow the reference architecture and standards. The OSI model shows how data and information travel through one software application (such as word processing software) to another executable file on a different network over a network media (including cable). The OSI reference model breaks down the difficulty of transferring files between devices across a network into *SEVEN* simpler, more controllable challenges.

STP

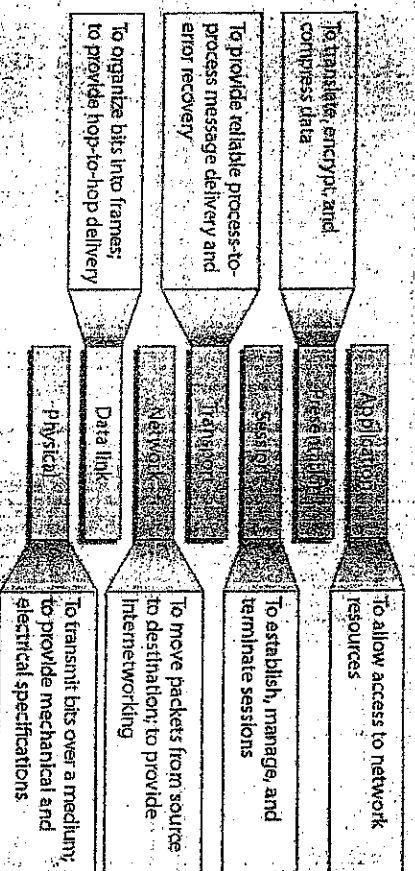


Figure 2.2: Open Systems Interconnection

The bottom four layers (transport, network, data link, and physical. Layers 4, 3, 2, and 1) are focused only on the transfer of data from one end of the connection to the other. The top four levels of the OSI model (application, presentation, and session—layers 7, 6, and 5) are primarily focused on software solutions. As data is transferred down the layers before network transport, it is wrapped with the required protocol details.

2.4.1 Physical Layer

The physical layer is the last bottom layer of the OSI architecture, and it organizes the actions needed to send a bit stream through physical media. It also specifies the processes and strategies that hardware objects and interfaces must carry out for transfer to take place. The physical layer defines the physical, electrical, operational, and organizational requirements for accessing information network connectivity, as well as the kind of transmission medium and communication scheme (simplex, half duplex, or full duplex).

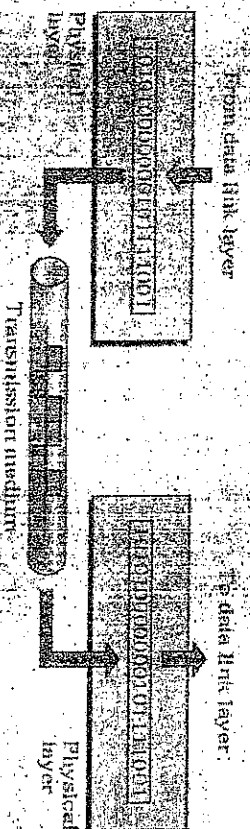


Figure 2.3 Physical layer

Copper cable, fiber-optic cable, and radio communications propagation are examples of transmission channels specified by the physical layer. The carrier system used to transport digital signals across network sites is also part of the fundamental layer. Carrier systems are basic data transmission systems that use copper or optic fiber optic cables, as well as wireless setups like microwaves, satellites, and mobile radio systems, to transport information.

2.4.2 Data-link layer

This layer is accountable for node-to-node transfer and turns the physical layer, which is a raw transmitting infrastructure, into a dependable link. It gives the top layer the impression that the physical layer is error-free (network layer).

Data from the physical layer is packaged into units known as blocks, frames, or packets by the data link layer. The data link layer includes a header to the frame to identify the physical sender address (source address) and/or recipient (destination address) if the frame is to be sent to other computers within the network. Flow control, security systems, and error detection and correction are all provided by the data-link layer.

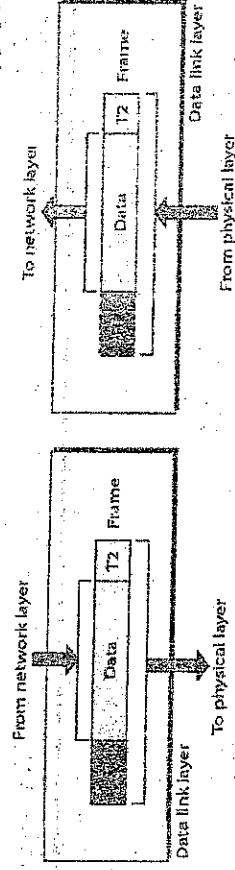


Figure 2.4 Schematic diagram of a Data Link layer

2.4.3 Network Layer

Data may be transported across equipment in a system utilizing different networks, subnetworks, or both courtesy of the network layer. This is in charge of addressing messages and data to ensure that they are delivered to the proper location, as well as transforming logical addresses and names (such as FLAME) into physical addresses. This layer is also in charge of determining a routing path to the target computer.

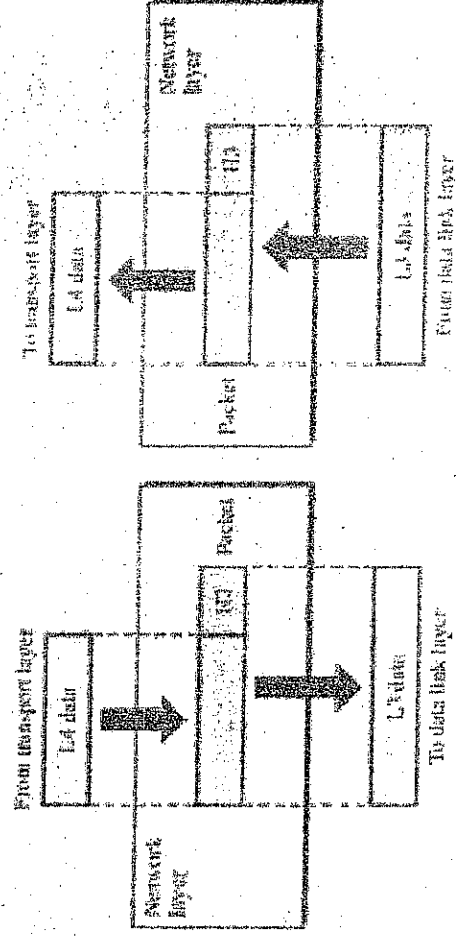


Figure 2.5 Network Layer

The network layer shields the hierarchy's higher levels from the data transfer and switching technologies that are used to link devices. Routers and their applications are connecting devices that function at the network layer.

2.4.4 Transport Layer

The transport layer manages and protects the end-to-end validity of the data packet as it travels via the link between two devices, allowing for secure and efficient data transmission. Data forwarding, sectioning, error correction, and two sorts of essential services to upper-layer guidelines: link-centric and connectionless are all responsibilities of the transport layer.

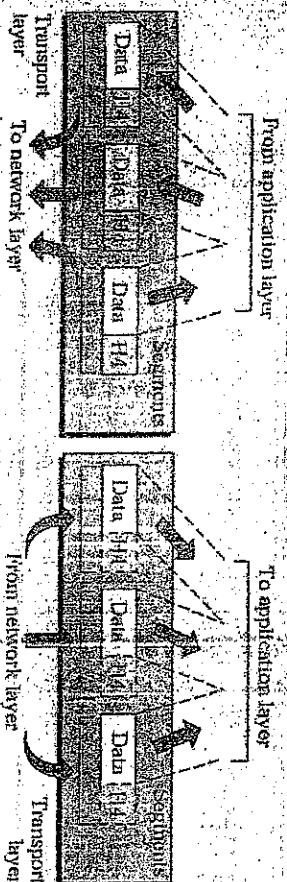


Figure 2.6 Schematic diagram of a Transport layer for OSI

In terms of communication, the transport layer is the uppermost layer in the OSI architecture yet can offer data gathering, link flow management, data ordering, error detection, and program identifying and authentication.

2.4.5 Session Layer

The dialog controller, also known as the session layer, offers a means for directing the exchanges between the two end devices. It specifies how to initiate, regulate, and terminate interactions (referred to as sessions) among programs.

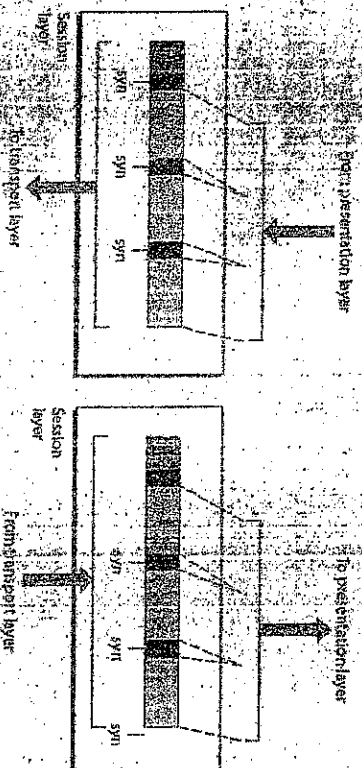


Figure 2.7 Schematic diagram of a Session layer of OSI

Just at the application layer, session layer protocols offer the logical connection elements. Data transfer protocols and email transmission are examples of these technologies. Connection sign-in and sign-back processes, as well as identity and access, are all part of the session's obligations. Remote connectivity across programs and entities, integration of data stream for retrieval reasons, the establishment of conversation sections and action units, connecting attribute adjustment, and dividing operations into subunits are all features of the session layer.

2.4.6 Presentation Layer

This layer gives application processes flexibility by dealing with any script or syntax translation required to convey data over the network in a standard communications file type. It describes how information should be formatted by end-user programs.

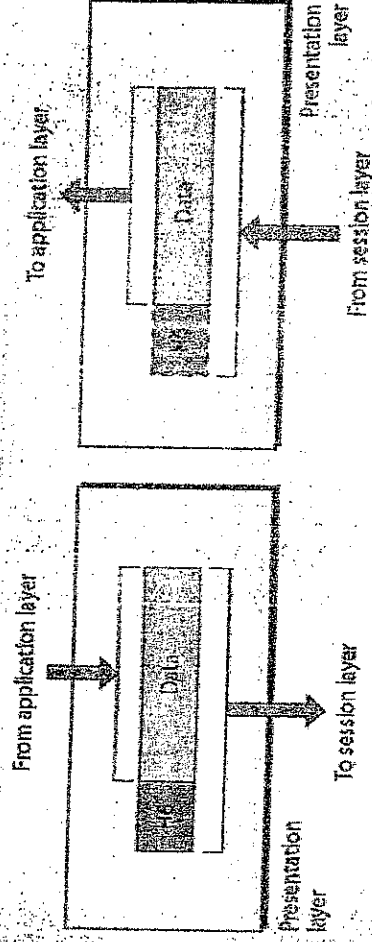


Figure 2.8 Schematic diagram of a Presentation layer of OSI

The presentation layer was responsible for translating across various data types and interfaces. Data file formatting, encrypting, encoding, and decoding of incoming packets, conversation processes, data compression methods, integration, disruptions, and discontinuation are all examples of presentation responsibilities.

2.4.7 Application Layer

The application layer, which connects directly to the GUI environment, is indeed the uppermost layer in the hierarchical model and is equivalent to the network's general manager. The applications layer offers distributed information capabilities and manages the chronology of activities inside a program as well as between the computer application and also another application's user.

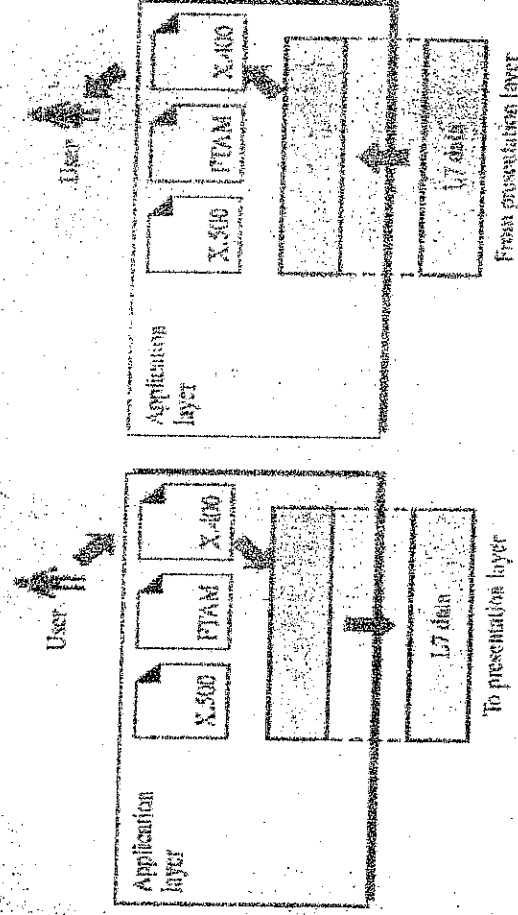


Figure 2.9 Schematic diagram of an application layer of OSI

The subscriber's application software interacts straightforwardly with the application layer. To engage the network infrastructure, user application procedures involve application layer service features. There are two types of operating elements: CASEs (common application service elements) that address specific demands of procedures such as connection management, synchronization, and retrieval. SASE (specific application service elements) is the second class, which includes TCP/IP stack, FTP, SNMP, Telnet, and SMTP.

2.5 Devices for Transmitting Data

A digital communications circuit's basic goal is to have a transmission line between destinations and to convey digital data through one unit (node, which houses computer systems or other telecommunication devices) to another through electrical circuits. Data communications circuits interface digital personal computers by utilizing digital communications equipment and facilities. Communication facilities are practical ways of linking hubs that are made available to data communications clients via public telephone networks (PTN), public data networks (PDN), and a variety of personal data transceivers.

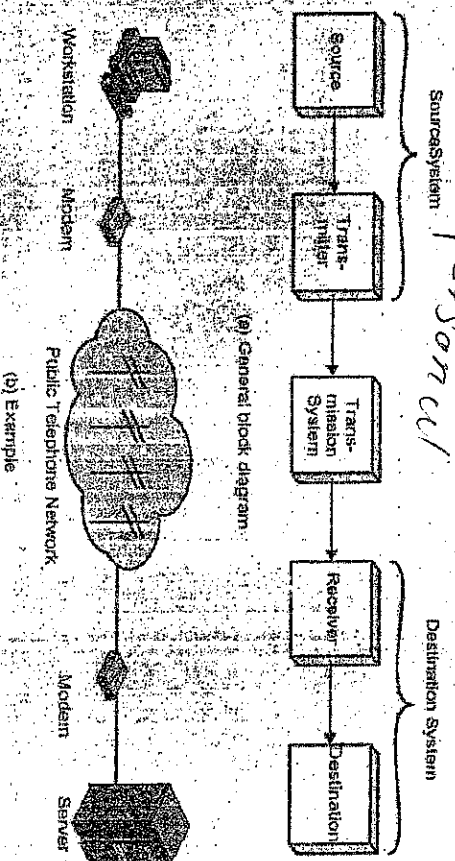


Figure 2.10 A Simple Two Station Data Communication Circuit

The diagram above depicts a simple two-station data communications circuit. The primary elements are as follows:

- **Source:** Such equipment creates the data that will be transferred, instances include a personal computer, a laptop computer, a workstation, etc. The source device enables users to input information into the system.
- **Transmitter:** A transmitter processes and encrypts data in order to generate electromagnetic signals that may be transferred through a transmission network. A modem, for instance, receives a digitized binary sequence out of a connected device such as a desktop pc, and converts this into an analog signal which can be managed by the telecommunications network.
- **Transmission medium:** This is responsible for transporting encoded signals from the transmitter to the receiver. Free-space radio communication (i.e., various categories of wireless transmission) and infrastructures such as copper and optical fiber cables are examples of communication mediums.
- **Receiver:** The receiver takes input from the communication links and translates this into a format that the destination node can understand. A modem, for instance, will take an analog signal from a connection or transmission line and then transform it into a digitized bit.
- **Destination:** Accept input data from the receiver which can be any form of digital hardware, such as the sources.

2.6 Data Transmission in Serial and Parallel Modes

There are two ways to send digital data: parallel and serial communications. In parallel data transmission, all bit of a binary code is transmitted concurrently. For instance, eight transmission lines are necessary to send 8-bit binary information through one unit

to the other. Every bit needs its data route. A word's bits are all conveyed in the same instance.

The technique of transferring binary texts one bit at a time is known as **serial data transmission**. Because the bits use the communication path in real-time, just one interconnection link is necessary. While serial data transmission is significantly cheaper and simpler than parallel data transmission due to the usage of one connecting line, it is a relatively sluggish mode of information transmission. In situations whereby fast speeds are not required, serial data transfer is appropriate.

Parallel communication is being used for short-distance data transfers within a pc, whereas serial transmission is being used for long-distance data transfers.

2.7 Circuit Topologies for Transmitting Data

A data communications circuit could be characterized in terms of the circuit configuration and transmission method.

2.7.1 Circuit Configurations

Data communications networks may be divided into two types: two-point and multipoint. A two-point configuration is made up of only two sites or units, while a multipoint system is made up of three or even more.

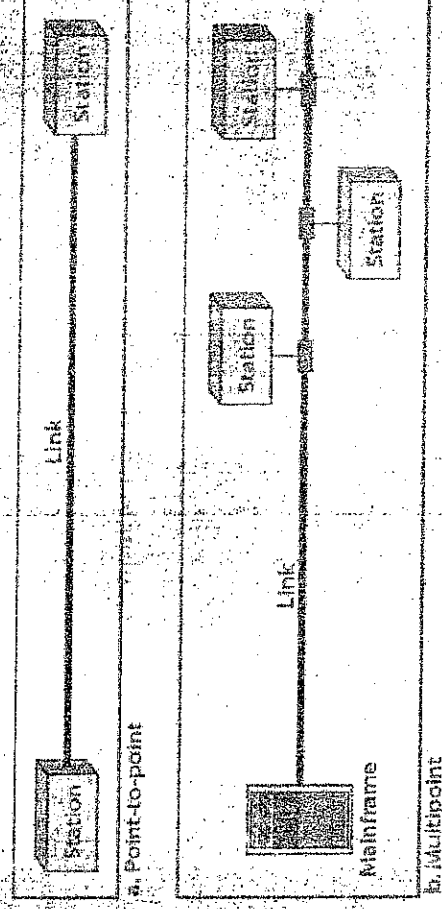


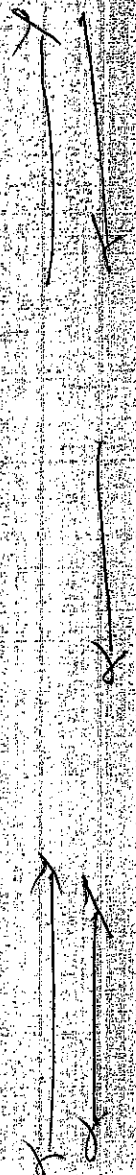
Figure 2.11 Circuit Configurations

A two-point circuit includes the exchange of digital data between one computer system and a desktop pc, two mainframe computers, or two different data communication systems.

A multi-point connection is commonly used to connect one mainframe computer (host) to several pcs or to connect many personal computers where the bandwidth is either Geographically distributed: Computers can utilize the connection concurrently or Subscribers alternate taking turns.

2.7.2 Transmission Modes

With data communications circuits, there are four transmission formats:

- 
- **Simplex Mode (SX):** This communication is unidirectional. On a connection, just one of the two devices can send, while the other can only receive. A good illustration is commercial radio transmission. Receive-only, transmit-only, and one-way-only links are all terms for simplex lines.

- **Half Duplex System (HDX):** Non-simultaneous bidirectional communication is possible with half-duplex radio systems that employ half-duplex radio frequencies. The first user can connect with the second user, but the second user can only speak with the first when the first has concluded his chat. At any one moment, the individual can only send or collect information. A walkie-talkie is a form of half-duplex device.

- **Full Duplex System (FDX):** Full-duplex systems support two concurrent connectivity. Both users can converse with each other at the same time. This may be accomplished by giving both users two concurrent but distinct channels.

2.8 Networks for Data Communications

A data communications network is any set of computers that are linked collectively, and networking is the act of exchanging resources between computers across a data network infrastructure. A data communications network's most critical aspects are efficiency, network throughput, dependability, and integrity.

2.9 Characteristics Properties, Operations, and a Network

End units, programs, and a connection that will handle communication between both the end stations are the major features of a connection. Servers, clients, communication channels, exchanging data, shared printers, and other accessories, software, and hardware resources, network interface card (NIC), local operating system (LOS), and network operating system (NOS) are all common equipment, services, and properties in computer systems.

- **Servers:** Servers are computers that store shared information, applications, and the system software for a connection. Servers provide all network users with access to network resources, and there are several types of servers. File servers, print servers, mail servers, communication servers, and other types of servers are illustrations.
- **Clients:** Clients are pcs that connect to and use a network and its services. Client computers are the network's clients (users), requesting and receiving services from the servers.
- **Shared Data:** Shared data refers to information that file servers make available to clients such as required data, printer access programs, and e-mail.
- **Shared Printers and other peripherals:** These are hardware resources made available to network users by servers. Datasets, printers, programs, or any other objects utilized by network users are examples of resources offered.
- **Network interface card:** Each node in the network does have a network interface card (NIC), a unique expansion card that organizes and delivers data, receives information, and manages the flow of data between the device and the network. The NIC transfers frames of data to the physical layer when transmitting, and also the NIC analyzes bytes retrieved from the physical layer and analyzes the signal depending on its characteristics on the receiving side.

- **Local operating system:** This enables personal computers to access files, print to a local printer, and utilize one or more storage media and CD drives. MS-DOS, UNIX, Windows XP, Vista, 10, and Linux seem to be a few instances.
- **Network operating system:** The network operating system (NOS) is software that runs on computers and servers and makes it easier to connect via a network. Clients can use the NOS to gain log-in abilities, password protection, printer access, network management activities, and information document sharing.

2.10 Models of Networks

Computer networks are divided into two types: peer-to-peer client/server networks and dedicated client/server networks. The client/server approach defines how two computers can connect with programs across a connection.

2.10.1 Client/server peer-to-peer network

All of the computers within the network share their information, such as disk drives and printers, with all the other computers on the network. Personal resources, like as hard disks, CD-ROM drives, and printers, are converted into shared, common resources available from any PC. With the exception of client-server networks, in which connectivity/data is concentrated on a file server Computer and made publicly available to multitudes of client Computers, peer-to-peer connection data is distinctively distributed. Although peer-to-peer Computers individually have their very own hard disk drives which are reachable to every computer, every PC functions both as a client (data requestor) and a server (information provider). If there are less than Ten clients on the network, protection is not really a problem, and all clients are in the same geographic region, a peer-to-peer network is a suitable solution.

The benefits of peer-to-peer networks over client-server network operating systems include the following:

- i. There is no requirement for a network manager.
- ii. The network is simple and affordable to configure.
- iii. For safety, every Computer can create duplicates of its information for all other Computers.
- iv. Peer-to-peer networks are the simplest to set up and are ideal for including both home and workplace use.

2.10.2 Dedicated client/server network

In this case, one pc serves as the server, while the others function as clients. By deploying one server for each program that resides inside an organization, dedicated server design can enhance the capacity of server-based networks. The selected servers hold all of the network's shared data and application software and act solely as a server, not as clients or workstations. Client PCs can connect to the servers and receive information and documents across the communication medium. Client computers in certain client/server networks send tasks to one of the servers, which perform the jobs and provide the output to the client computer.

For general-purpose data networks, the dedicated client/server paradigm is preferred to the peer-to-peer client/server approach.

2.11 Architectures of networks

Topology in networking refers to the arrangement of linked devices, i.e. how pcs, cables, or other peripherals of an information network infrastructure are visually and conceptually coupled. The logical topology defines how information travels across the network, whereas the physical topology specifies how the connectivity is really built out. Point-to-point and multipoint are the two most fundamental topologies. For high-speed digital content, a point-to-point architecture often connects two desktops. The star, bus, ring, mesh, and hybrid topologies are illustrations of multipoint topologies that interconnect three or more terminals using a common transmission channel.

2.1.1.1 Star topology

Each node (server, desktops, and accessories) in a star architecture is physically connected to the main network hub, switch, or collector. A star network's data is transferred via hub, switch, or collector prior to reaching its destination. All network functions are managed and controlled by the hub, switch, or collector. This also serves as an information flow repeater.

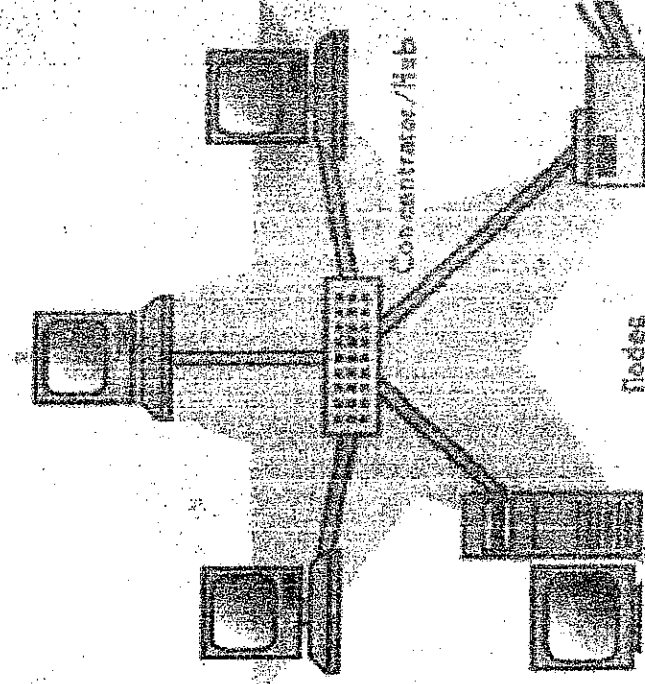


Figure 2.12 Node Connections in Star Topology

Table 2.1: Advantages and Disadvantages of Star Topology

Advantages	Disadvantages
Easily extended without interfering with the connection	More network Cables are required
A single member is affected by cable disruption.	A central linking device eliminates the possibility of a single-point failure.
Challenges are simple to diagnose and pinpoint.	Implementation is much more complex.

2.11.2 Bus topology

To interconnect all gadgets in a bus network, a single core is used. A single link (the backbone) serves as a common medium of communication to which devices directly link or tap via an interaction connection. A device wishing to interact with some other node in the network broadcasts a message over the channel, which all other devices view, and only the intended beneficiary receives and analyzes the information.

The bus topology is the most basic and widely used means of linking pcs. The communication line's two ends never contact to make a full circle. A bus topology is sometimes referred to as a multidrop, linear, or horizontal bus.

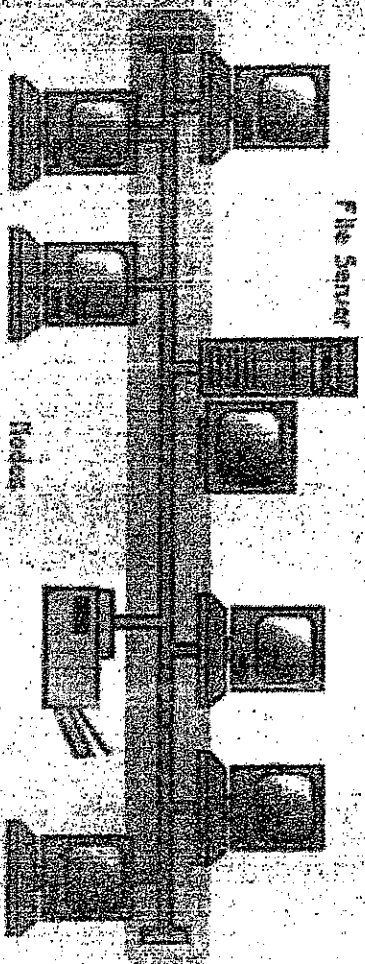


Figure 2.13 Node Connections in Bus Topology

Table 2.2: Advantages and Disadvantages of Bus Topology

Advantages	Disadvantages
Low-cost and simple to deploy	Whenever pcs are introduced or disconnected from a network, it causes disturbance.
A minimal network cable is needed.	All computer systems will be unable to connect to the network if the cable breaks.
It makes no use of any specialist network hardware.	Troubleshooting is challenging.

2.11.3 Ring topology

Each unit in a ring network (also referred to as a loop) has precisely two neighbors for purpose of communication. All signals go in the same manner across a ring (either "clockwise" or "counter-clockwise"). Every one of the terminals is linked together in succession to make a complete ring or circle. Communications are routed via all of the units in the circle and have to do so. The ring network is comparable to bus or star topologies in that each computer functions as a repeater.

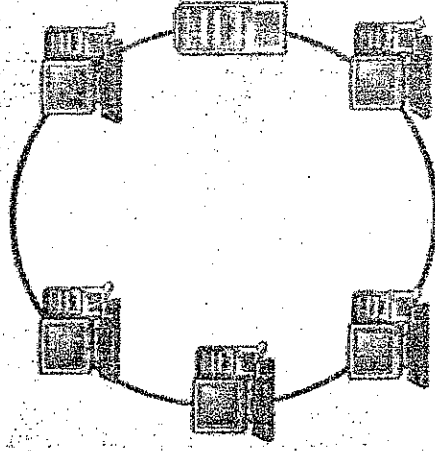


Figure 2.14 Node Connections in Ring Topology

Table 2.3: Advantages and Disadvantages of Ring Topology

Advantages	Disadvantages
Cable faults are quickly identified, enabling maintenance simpler.	Network extension may result in network interruption.
Ring networks are reasonably simple to set up.	A single cable failure may bring the entire network down.

2.11.4 Mesh topology

The mesh topology integrates a one-of-a-kind network architecture whereby each computer on the network links to all the others, resulting in point-to-point connectivity among all network nodes. The information delivered on a mesh network, unlike with the preceding topologies, can traverse any of the multiple potential pathways from

sender to receiver. A complete mesh network is one where each device connects to every other device.

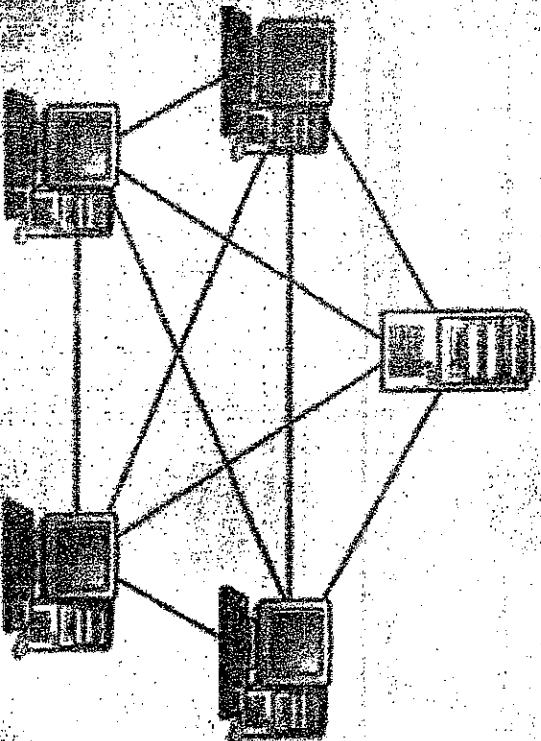


Figure 2.15 Node Connections in Mesh Topology

Table 2.3: Advantages and Disadvantages of Mesh Topology

Advantages	Disadvantages
Allows for an alternative route across devices.	It needs extra wire than other LAN topologies.
Current users will not be inconvenienced if the network is extended.	Deployment is difficult.

2.1.1.5 Hybrid topology

This topology (also known as mixed topology) is basically the combination of two or even more standard topologies to create a bigger, greater complicated topology. The primary purpose is to convey the benefits of various topologies.

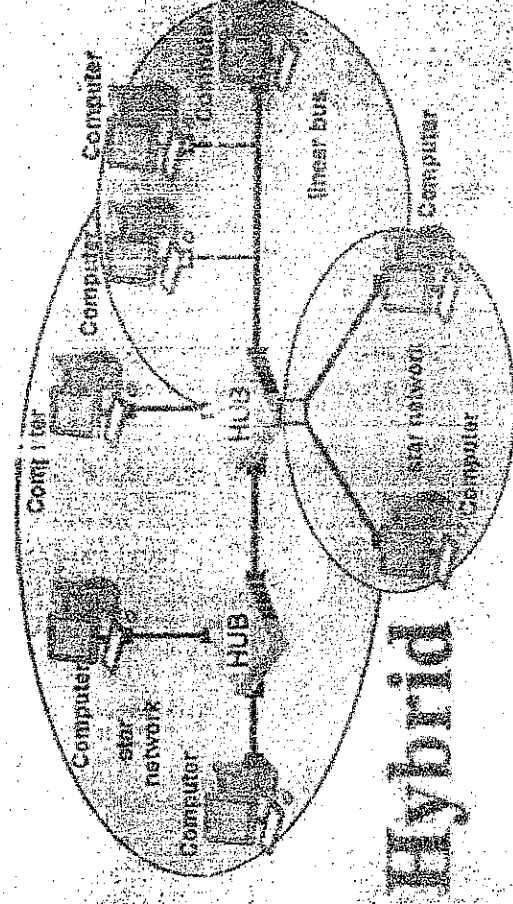


Figure 2.16 Hybrid topology

2.12 Categorizations of Networks

The extent or size of a computer network infrastructure is one approach to categorizing the many forms of computer network architectures. The following are possible examples of area network types:

2.12.1 Local area network

A local area network (LAN) is a network that links pcs within a certain geographical region, such as a house, classroom, digital library, office complex, or cluster of houses. A network operating system is used by LANs to offer two-way communications at bit rates ranging from 10 Mbps to 100 Mbps. LANs are often owned, operated, and managed by one person individual or group, in addition to functioning in a small location. They also have a preference for particular networking technologies, most notably Ethernet and Token Ring.

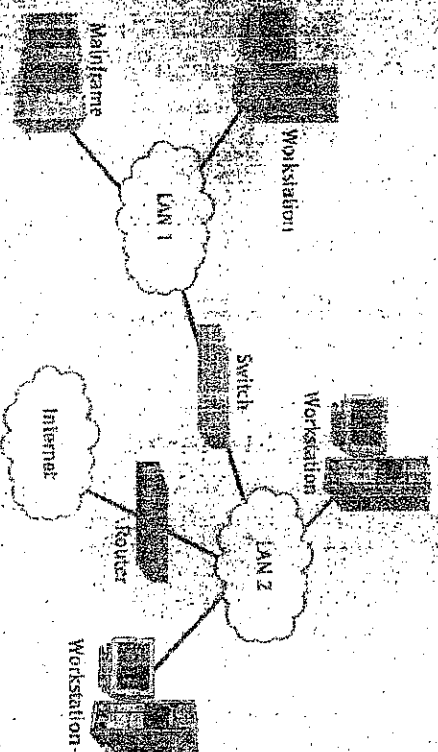


Figure 2.17: A LAN that connects to another LAN, the Internet, and a desktop PC
Benefits of LAN

1. Allocate resources judiciously.
2. Single workstations may be capable of surviving network disruption if they are not dependent on others.
3. Hardware evolution is distinct from system evolution.
4. Transmission rates are high while error rates are very low.
5. Compatibility for a wide variety of hardware and software

2.12.2 Metropolitan area network

A MAN is designed to cover a broader landmass than a LAN, spanning from a few blocks of buildings to a whole metropolis. Its geographical coverage sits in between a WAN and a LAN. A MAN can be a single network, such as a cable television network, or it can join several local area networks (LANs) utilizing high-capacity backbone technologies, such as fiber-optical cables, and offer up-link services to wide area networks and the Internet. MANs generally run at speeds ranging from 1.5 Mbps to 10 Mbps and with lengths ranging from 5 miles to a few hundred kilometers. FDDI (fiber distributed data interface) and ATM are two instances of MANs (asynchronous transfer mode).

2.12.3 Wide area network

Wide area networks (WANs) are the most fundamental type of information communications network, providing very slow-speed, long-distance transfer of data, audio, and visual content over significantly larger and widely distributed geographical areas, such as a nation or a continent. WANs connect routers in various places. A WAN is distinct from a LAN in numerous respects. Most WANs (including the Internet) are collectively or distributed owned and managed rather than owned and managed by a single company. For long-distance communication, WANs often employ technologies such as ATM, Frame Relay, and X.25.

2.12.4 Global area network:

A global area network (GAN) connects nations all over the world. The Internet is a fantastic example since it is fundamentally a network made up of various networks that link nearly all countries on the planet. GANs function from 1.5 Mbps to 100 Gbps and cover thousands of miles.

Exercises

- 1) With the aid of a block diagram explain radio transmission links.
- 2) Describe the five (5) basic parts that make up a digital communication system.
- 3) State and give a brief explanation of Data Communication Standards Organization you know.
- 4) Give the seven layers Open Systems Interconnection (OSI) and their corresponding characteristics.
- 5) Explain the concept of Data Transmission in Serial and Parallel Modes.
- 6) Describe the transmission Modes and characteristic properties of network for data communication.
- 7) State and explain the Basic architectures of network.

